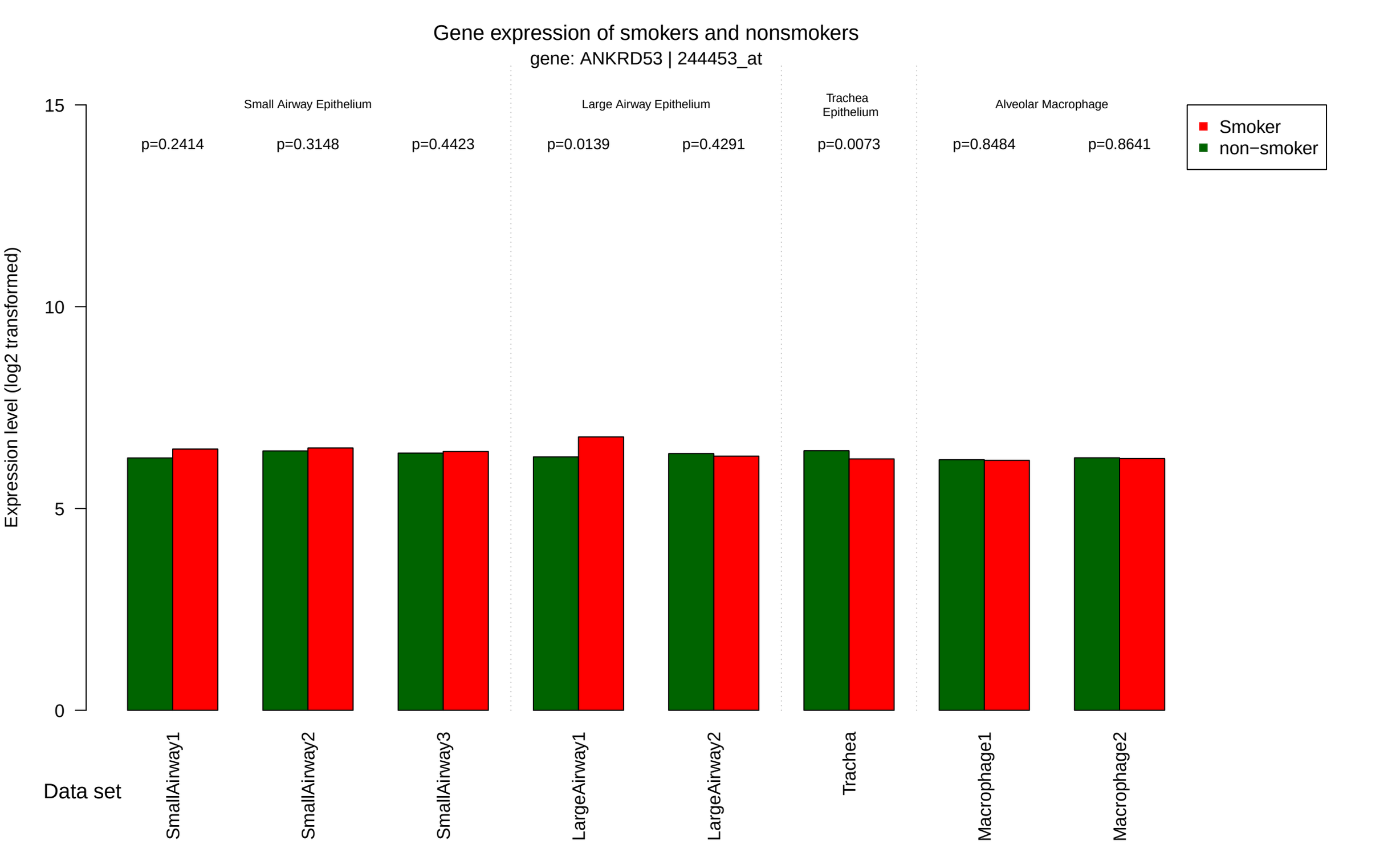
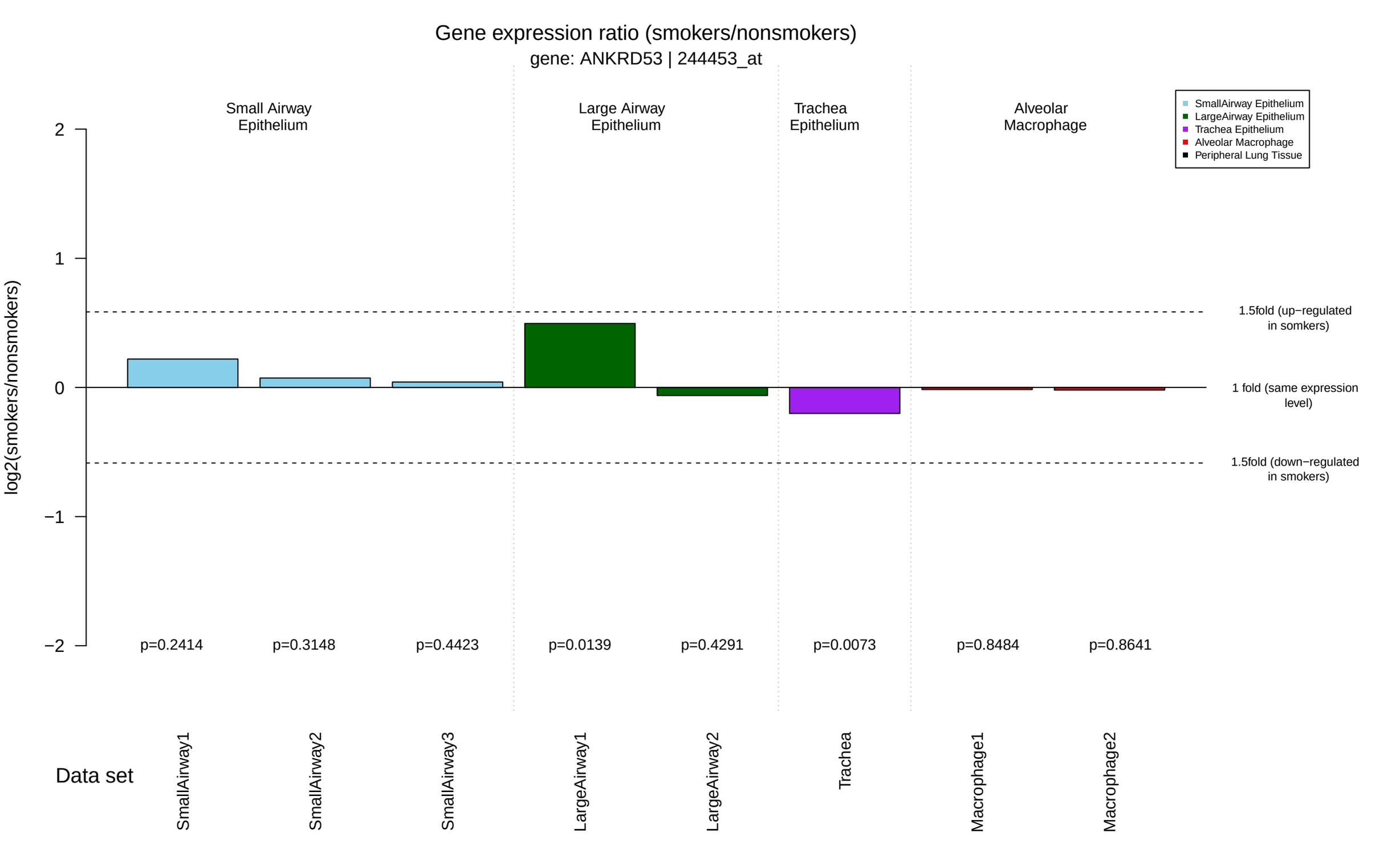
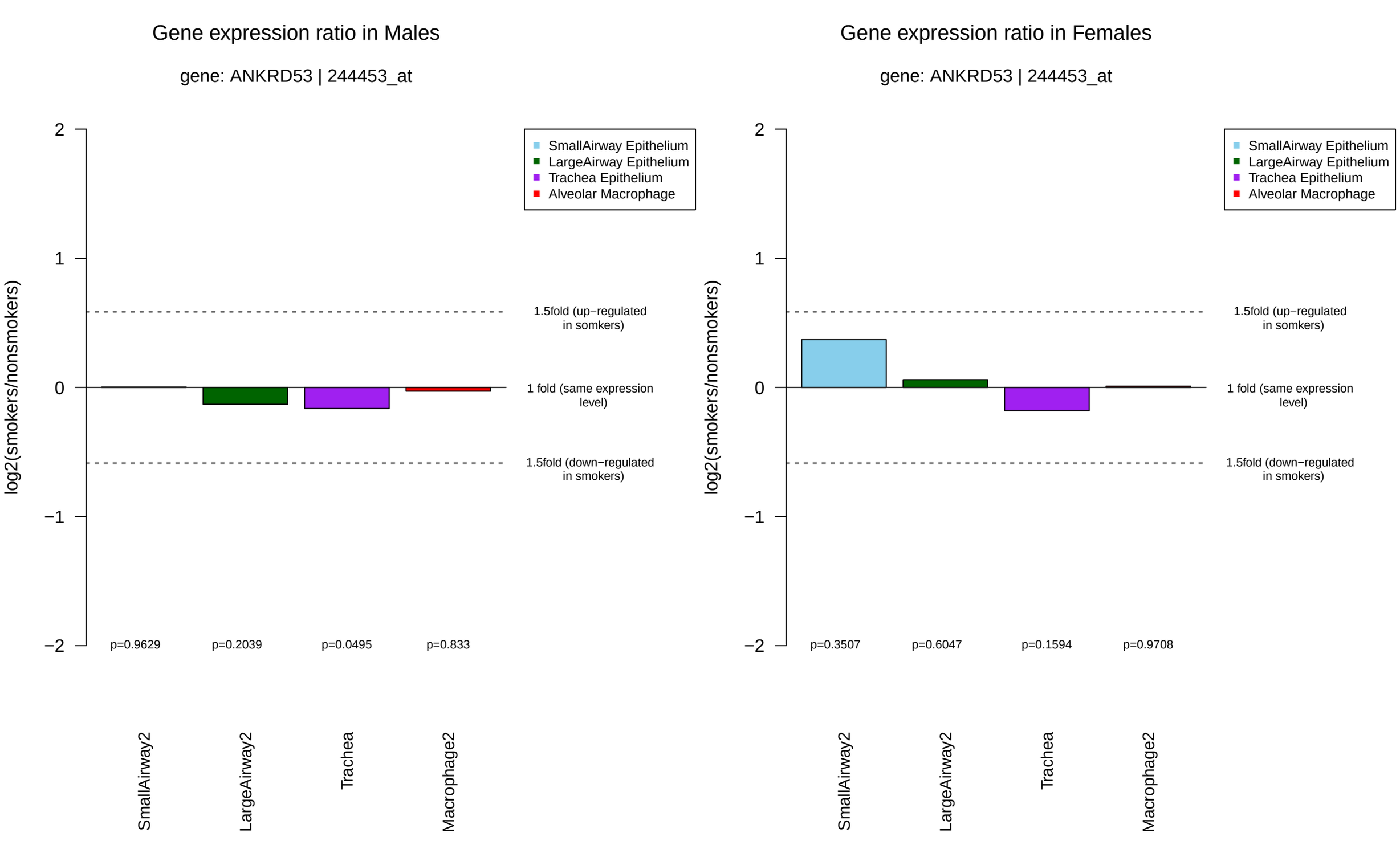
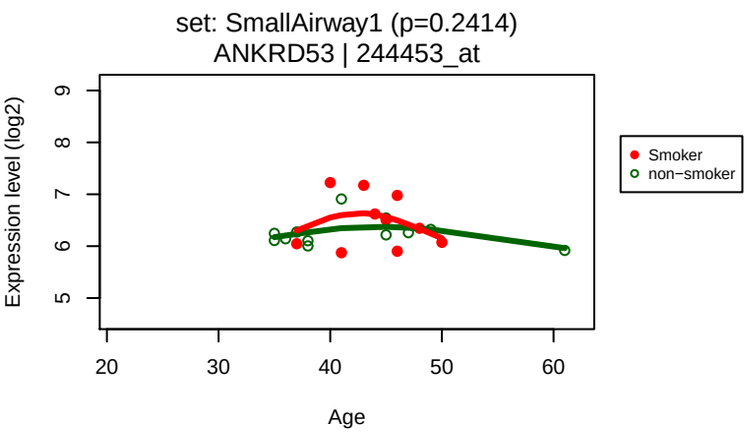
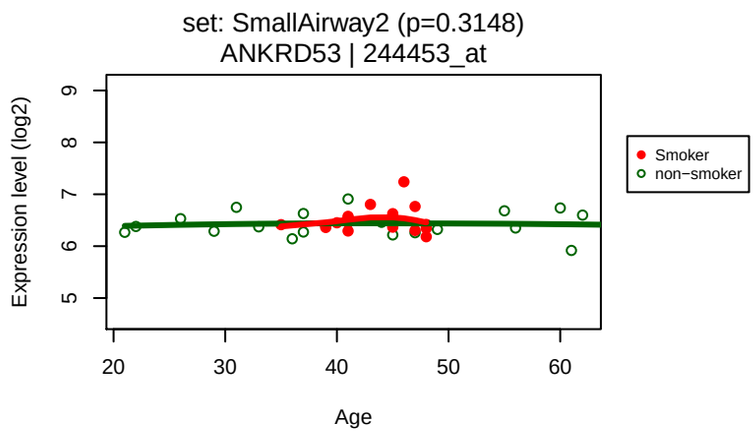
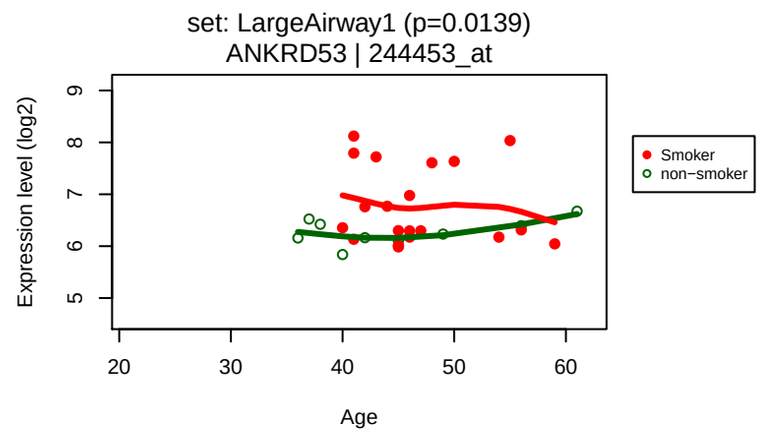

Gene expression of smokers and nonsmokers
gene: ANKRD53 | 244453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2414
p=0.3148
p=0.4423
p=0.0139
p=0.4291
p=0.0073
p=0.8484
p=0.8641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

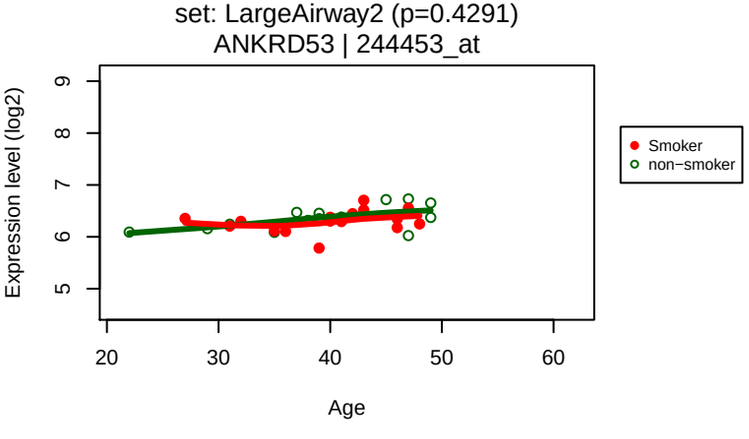

Gene expression ratio (smokers/nonsmokers)
gene: ANKRD53 | 244453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2414
p=0.3148
p=0.4423
p=0.0139
p=0.4291
p=0.0073
p=0.8484
p=0.8641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

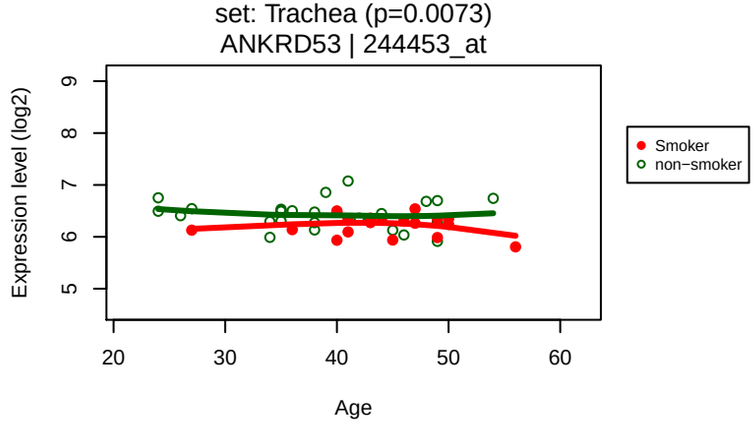

Gene expression ratio in Males
gene: ANKRD53 | 244453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9629
p=0.2039
p=0.0495
p=0.833
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANKRD53 | 244453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3507
p=0.6047
p=0.1594
p=0.9708
SmallAirway2
LargeAirway2
Trachea
Macrophage2

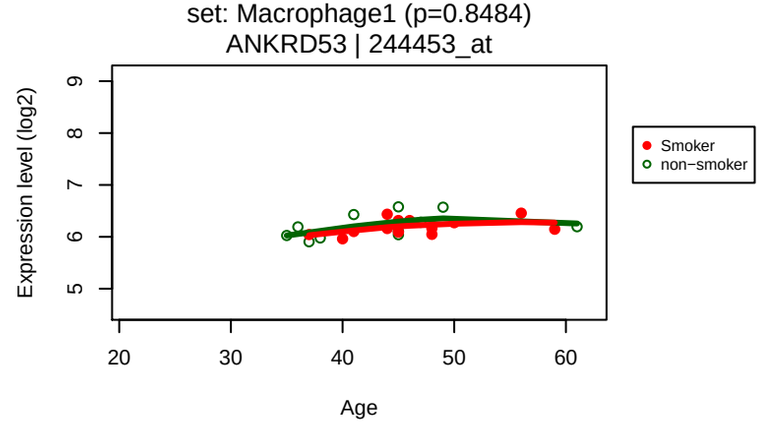

set: SmallAirway1 (p=0.2414)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker

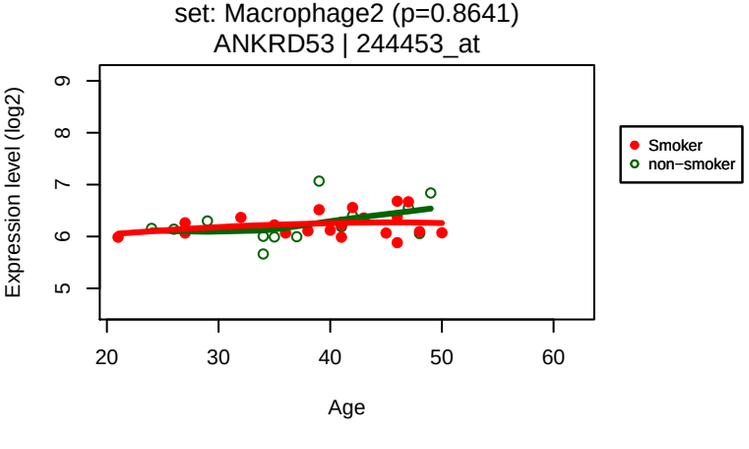

set: SmallAirway2 (p=0.3148)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker

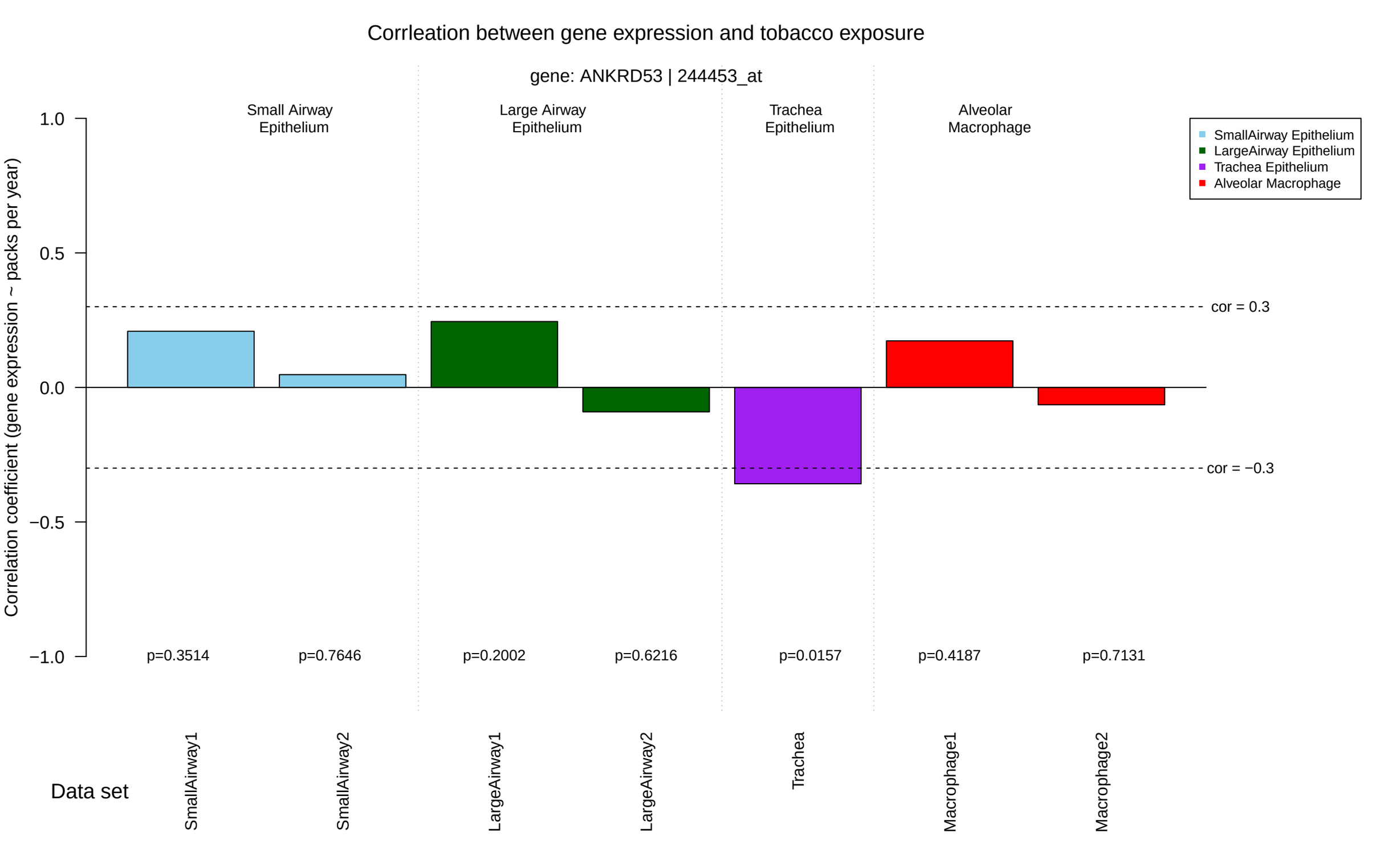

set: LargeAirway1 (p=0.0139)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4291)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0073)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8484)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8641)
ANKRD53 | 244453_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANKRD53 | 244453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3514
p=0.7646
p=0.2002
p=0.6216
p=0.0157
p=0.4187
p=0.7131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2